

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. BFPCL-DSV-RAEC 2026

Project	FEED for Gas Gathering Pipelines, Gas Processing Plant & Methanol Plant
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Originating Company	Epic Atlantic Limited
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Produced Water Handling, Processing, & Disposal Philosophy

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Rev No.	Date	Description of Revision
R01	07 Mar 2026	For Review by RAEC, and Subsequent Discussion with NUPRC, in respect of BBOU & ELEP FDP
R02	25 Mar 2026	Clarification that clearly brings out BFPCL options plan without ambiguity
R03	09 May 2026	Updated to include Zero Liquid Discharge and River Water Intake
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer and Petrochemical Company Limited	
Project:	FEED for Gas Gathering Pipelines, Gas Processing Plant & Methanol Plant	Project No. BFPCL-DSV-RAEC 2026
Facility:	Gas Gathering Pipelines, Gas Processing Plant & Methanol Plant	Doc No. EA-BFPCL-DSV-GGPL-FEED-GEN-PHY-004

This document demonstrates BFPCL/DSV Management plan and commitment to follow the regulatory & policy framework towards produced water handling, treatment and disposal. To this effect BFPCL/DSV has developed a Plan, including Framework, Road Map and Decision Flow Charts, Benchmarks and Best practices, for systems, procedures, and controls for safe handling, treatment, monitoring, and disposal of produced water generated in the project.

Six water handling, treatment and disposal options are considered, The preferred strategy for the AKAB Gas Hub is the **Industrial Re-use** of produced and process water. Under this model, water from all sources is routed to a centralized **Water Treatment Plant (WTP)** at the Akabel (AKAB) Hub to refine the water to "Electronic-Grade" purity, ultra-pure feedstock to the Methanol Plant.

At peak operations, the project transitions toward **Zero Liquid Discharge (ZLD)**. While the initial internal water supply (formation water, GPP condensation, and MP reaction byproducts) is estimated at **56,5 MBWPD**, the peak demand of the 10,000 MTPD Methanol Plant is approximately **82 MBWPD**. The balance of **25,5 MBWPD** is sourced via a dedicated **St Nicholas River Water Intake Skid (RWIS)**, providing absolute supply redundancy.

BFPCL/DSV is committed to full regulatory compliance in alignment with the reference documents below:

The Water Treatment will start with 3-phase Separator, followed by a cascade of downstream treatment units designed to progressively remove oil, suspended solids, and fine particulates.

REFERENCES	
Document No.	Description
EA-BFPCL-GMMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	Petroleum Industry Act (2021)
	EGASPIN (administered by NUPRC)
	EIA Act
	National Oil Spill Detection and Response Agency (NOSDRA Act)
	National Environmental Standards and Regulations Enforcement Agency (NESREA Act)

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Abbreviations

Abbreviation	Meaning
AKAB	Akabel
AOP	Advanced Oxidation Process
API	American Petroleum Institute
ASU	Air Separation Unit
ATR	Autothermal Reformer
BBOU	Bubouwe Bou Field
BFPCL	Brass Fertilizer and Petrochemical Company Limited
BOD	Biochemical Oxygen Demand
BTEX	Benzene, Toluene, Ethylbenzene, and Xylene
BWPD	Barrels of Water Per Day
CAP	Corrective Action Plan
Cd	Cadmium
COD	Chemical Oxygen Demand
COREN	Council for the Regulation of Engineering in Nigeria
CF	Cartridge Filter
CPF	Central Processing Facility
Cr	Chromium
DAF	Dissolved Air Flotation
DGF	Dispersed Gas Flotation
DMF	Dual Media Filter
DSV	DSV Pipetronix Limited
DW-01	Disposal Well 01
ECMR	Environmental Compliance Monitoring Report
EA	Epic Atlantic (Document Prefix)
EGASPIN	Environmental Guidelines and Standards for the Petroleum Industry in Nigeria
EIA	Environmental Impact Assessment
ELEP	Elepa Field
EOR	Enhanced Oil Recovery
EPF	Early Production Facility

Abbreviation	Meaning
ESG	Environmental, Social, and Governance
FDP	Field Development Plan
Fe	Iron
FEED	Front-End Engineering Design
GGPL	Gas Gathering Pipelines
GOSP	Gas-Oil Separation Plant
GPMC	Gas Processing & Methanol Complex
GPP	Gas Processing Plant
GSPA	Gas Supply and Purchase Agreement
HAZOP	Hazard and Operability Study
HSE	Health, Safety, and Environment
IGF	Induced Gas Flotation
ISO	International Organization for Standardization
IX	Ion Exchange
LOF	Life-of-Field
MBWPD	Thousand Barrels of Water Per Day
MEG/TEG	Monoethylene Glycol / Triethylene Glycol
MMscfd	Million Standard Cubic Feet per Day
MSDS	Material Safety Data Sheet
MTPD	Metric Tonnes Per Day
NESREA	National Environmental Standards and Regulations Enforcement Agency
NGL	Natural Gas Liquids
Ni	Nickel
NIWA	National Inland Waterways Authority
NOSDRA	National Oil Spill Detection and Response Agency
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OEMS	Online Effluent Monitoring System
OIW	Oil-in-Water
OKNU	Okpokunou Field
OML	Oil Mining Lease

Abbreviation	Meaning
Pb	Lead
PAH	Polycyclic Aromatic Hydrocarbons
PIA	Petroleum Industry Act
PWMP	Produced Water Management Plan
RAEC	Regulatory Affairs and Environmental Compliance
RCA	Root Cause Analysis
RENAISSANCE (RAEC)	Renaissance Africa Energy Company Limited
RO	Reverse Osmosis
RWIS	River Water Intake Skid
SDI	Silt Density Index
SBM	Single Buoy Mooring
SCADA	Supervisory Control and Data Acquisition
S/C	Steam-to-Carbon Ratio
SMR	Steam Methane Reformer
TDS	Total Dissolved Solids
TSS	Total Suspended Solids
TVD	True Vertical Depth
WTP	Water Treatment Plant
WTU	Water Treatment Unit
ZLD	Zero Liquid Discharge

1. Executive Summary

This document outlines BFPCL/DSV's management philosophy and commitment to comply with the applicable regulatory and policy framework governing produced water handling, treatment, and disposal.

Produced Water comes from 3 sources: 1) formation water from wet gas gathering wells run through a 125km pipeline network, 2) condensed water recovered during the gas dehydration and cooling processes within the 500 MMscfd Gas Processing Plan (GPP) and 3) "Reaction Water" generated as a byproduct of the methanol synthesis process of the 10,000 MTPD Methanol Plant (MP). Both plants are at the AKAB Terminal

Six water handling, treatment and disposal options are considered, namely:

- 1) Industrial Re-use (The Core Goal): Produced water is treated to ultra-high purity levels as steam feedstock for the **10,000 MTPD Methanol Plant**, Achieves Zero Liquid Discharge
- 2) Hybrid: Re-use + BR Transfer (18km): Produced water from AKAB Terminal piped 18km pipeline to the nearby **Brass Terminal** to dispose surplus water at reduced capex
- 3) Hybrid: OKNU Source Sep + Barge: Water & condensate separated upstream the 125km trunkline and moved by **barges** to the AKAB Terminal.
- 4) Hybrid: OKNU Source Sep + Spiking: Separated condensate/water re-injected into the gas for for multiphase flow in the 125km pipeline to AKAB Terminal
- 5) Offshore Outfall & Emergency Hybrids: Dedicated **22.2km subsea pipeline** leads from the AKAB Terminal to a deep-water diffuser to dispose the waters
- 6) 3rd Party Outsourcing: The liability for water disposal is shifted to a contractor, who collects, treats & disposes the water at their own government-approved facilities but at high opex.

The preferred strategy for the AKAB Gas Hub is the **Industrial Re-use** of produced and process water. Under this model, water from all sources is routed to a centralized **Water Treatment Plant (WTP)** at the Akabel (AKAB) Hub. This facility is equipped with an integrated "**Triple-Shield**" configuration—consisting of a **Methanol Stripper**, **Advanced Oxidation (AOP)**, and **Double Pass Reverse Osmosis (RO)** with **Ion Exchange** modules—to refine the stream to "Electronic-Grade" purity. This ultra-pure feedstock protects the sensitive **SMR + ATR** catalysts from carbonization and chemical poisoning, ensuring a >20-year operational lifecycle.

At peak operations, the project transitions toward **Zero Liquid Discharge (ZLD)**. While the initial internal water supply (formation water, GPP condensation, and MP reaction byproducts) is estimated at **56,5 MBWPD**, the peak demand of the 10,000 MTPD Methanol Plant is approximately **82 MBWPD**. The balance of **25,5 MBWPD** is sourced via a dedicated **St Nicholas River Water Intake Skid (RWIS)**, providing absolute supply redundancy.

To ensure resilience during non-steady-state periods—such as plant maintenance, single-train operations, or late-life high water-cut scenarios—surplus water is managed via a **22.2km Offshore Subsea Outfall**, regional terminal transfers, or marine shuttle exports. Furthermore, a **High-Capacity Finger Slug Catcher** is integrated at the AKAB hub to mitigate multiphase surges, ensuring a stabilized, continuous feed to the treatment units and the synthesis reactors.

BFPCL/DSV's objective is to ensure **full regulatory compliance and environmental stewardship**, in alignment with the following Nigerian statutory and regulatory instruments:

- **Petroleum Industry Act**
- **Environmental Guidelines and Standards for the Petroleum Industry in Nigeria** administered by the Nigerian Upstream Petroleum Regulatory Commission
- **Environmental Impact Assessment Act**
- **National Oil Spill Detection and Response Agency Act**
- **National Environmental Standards and Regulations Enforcement Agency Act**

The Produced Water management will be based on **regulatory compliance, environmental protection, operational reliability, lifecycle cost, and long-term field development strategy**.

2. Introduction

2.1 Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (AKAB) in Odioma, Brass LGA, Bayelsa state. The plant complex is to focus on phase-1 production of natural gas feedstock from 2 RAEC fields, - Elepa (ELEP) & Bubouwe Bou (BBOU) in OML 33, anchored on GSPA 270, and the import is the production from these two fields.

In as much as the focus is on BBOU & ELEP, we would like to draw our attention to Okpokunou (OKNU) production, anchored on a separate GSPA 200 which would invariably impact on produced water volumes, hence the water handling, processing, and disposal plan.

It is therefore of pragmatic imperative that Okpokunou production be included in the produced water handling, processing, and disposal plan in BBOU & ELEP, as the OKNU productions would be happening at the same time as BBOU & ELEP.

The Project involves the development, construction, and operation of a plant comprising:

Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol 500 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.

Gas gathering pipelines and associated manifolds – BBOU-ELEP-AKAB Pipelines, comprising a combination of 20" x 8km BBOU-ELEP Pipeline Segment + 28" x 32km ELEP-AKAB Pipeline, and the 20" x 85km OKNU-BBOU pipeline

Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.

All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

Produced water is generated from 3 sources, with initial production of 36 Mbwpd, increasing up to 40 Mbwpd over life of field (LOF).

Wells Crude Production: Elepa, and Bubouwe Bou, and Okpokunou crude production; and under a separate GSPA 200, from formation water. Initial water production will be 36 Mbwpd and will increase, as the field matures, up to, 40 Mbwpd over life-of-field (LOF).

Gas Processing Plant: Water knock out from 500 MMscfd from a temperature drop from 40°C to 15°C from propane refrigeration or JT valves & condensate water mix.

Methanol Plant: "Reaction Water" -byproduct of CO₂ hydrogenation in methanol production or syngas production

The produced water will be:

Treated to meet EGASPIN discharge limits administered by NUPRC.

Primarily re-injected into a designated disposal well, if available..

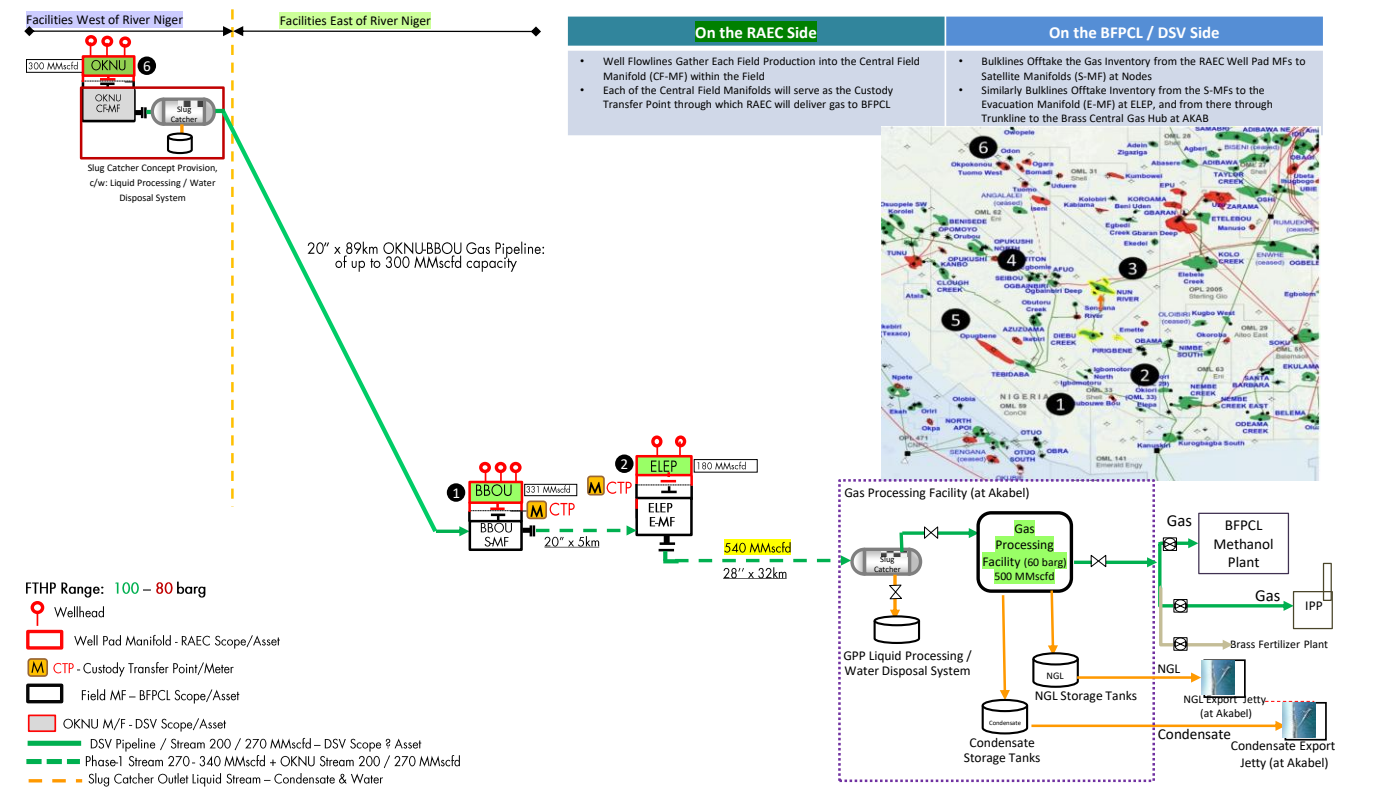
Surface discharged only under contingency conditions and within regulatory limits.

The treatment system will be designed with +30% capacity margin and redundancy to ensure continuous compliance. Online oil-in-water (OIW) monitoring and automatic diversion systems will be included.

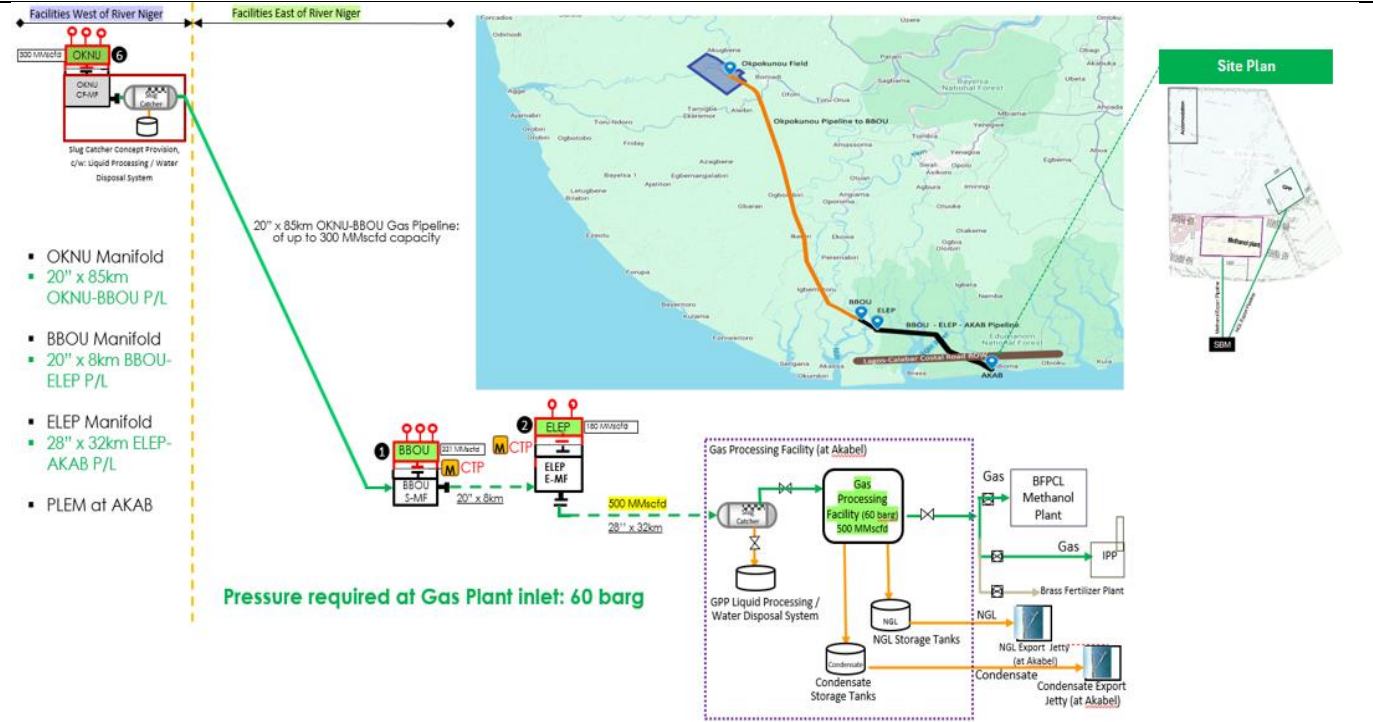
BFPCL commits to full regulatory compliance, environmental protection, and continuous monitoring in accordance with applicable Nigerian regulations.

2.2 Schematics of FEED Scope

Schematics: - Phase-1 Gas Supply from BBOU & ELEP alongside Okpokunou Gas Supply



Satellite Imagery: - Phase-1 Gas Supply from BBOU & ELEP alongside Okpokunou Gas Supply



3. Regulatory Standards and Requirements

3.1 Regulatory Instruments

Key governing instruments include:

- Petroleum Industry Act (PIA) 2021
- Environmental Guidelines and Standards for the Petroleum Industry in Nigeria (EGASPIN)
- NUPRC Upstream Petroleum Environmental Regulations
- Applicable DPR legacy guidelines (where referenced)
- National Environmental (Surface & Groundwater Quality Control) Regulations

3.2 Regulatory Standards / Limits

3.2.1. Disposal Methods in NUPRC Guidelines

The disposal methods recommended under the current NUPRC regulations are outlined below. All identified methods have been duly considered in the BFPCL produced water treatment, handling, and disposal study, and have informed the development of the primary disposal pathways evaluated:

Method	Regulatory Status	Requirements
Re-injection	Preferred	Water is pumped back into the reservoir for pressure maintenance. Requires NUPRC approval and integrity tests. • Into depleted reservoirs or designated disposal wells • Requires reservoir compatibility study • Must prevent formation damage and groundwater contamination • Continuous pressure monitoring required
Industrial Re-use	Preferred	• Approval of WTP Engineering Design & EIA • Inspection of Online Monitoring Systems (OEMS) • Monthly Effluent Quality Reports (EQR) • Environmental Evaluation Report (EER)
Surface Discharge (Onshore/Swamp Conditions)	Restricted	Discharging into freshwater rivers is increasingly discouraged. • Only after full treatment • Must meet EGASPIN effluent standards • Discharge point must be environmentally approved • Diffuser systems required in some cases
Offshore Discharge	Permitted	Allowed beyond a certain distance from the shoreline, provided oil content is within the 40 mg/L limit. • Online Oil-in-Water (OIW) analyzers installed that provide a continuous data feed to NUPRC and • an Automatic Diverter Valve triggered by the 40mg/L limit, sending the contaminated water back to a "slop tank" rather than into the ocean.

Evaporation Containment /	Limited Use	• Requires lined pits • No unlined earthen pits permitted • Must prevent seepage to groundwater
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3.2.2. Effluent Discharge Limits (2026)

Regulations differentiate strictly between **Inland/Nearshore** (high-risk swamp and land areas) and **Offshore** (deep sea) operations; the following limits will be observed:

Parameter	Inland / Nearshore	Offshore*
Oil & Grease	< 10 mg/L	< 30 mg/L
Total Dissolved Solids (TDS)	< 2,000 mg/L	Ambient Seawater levels
Chemical Oxygen Demand (COD)	< 125 mg/L	Monitoring required
BOD (Biol. Oxygen Demand)	< 30 mg/L	< 50 mg/L
Phenols	< 0.5 mg/L	< 1.0 mg/L
Methanol Content	< 2.0 mg/L (Residual)	< 5.0 mg/L
pH	6.5 – 8.5	6.5 – 8.5
Heavy Metals (Pb, Hg, Cr)	Near-zero / Non-detectable	Strictly monitored
Turbidity	< 10 NTU	N/A
Temperature	Ambient $\pm$ 3°C	Ambient $\pm$ 5°C

* >12 nautical miles / 22.2km

3.3 Regulatory Monitoring and Reporting Requirements

3.3.1. Statutory Operational Control

As a responsible corporate entity, BFPCL will implement the following requirements:

- *Environmental Management Plan (EMP)*: will be prepared and approved to demonstrate how produced water will be minimized and treated.
- *Environmental Risk Register (ERR)*: of potential water-related risks (e.g., pipe bursts, treatment failure) will be maintained.
- NUPRC Quarterly *Effluent Monitoring & Reports (ECMR)*:
- Produced water management plan (PWMP)
- The Environmental Remediation Fund (ERF) from Section 103 of the PIA, all operators must contribute to enable the NUPRC to clean up the site immediately.
- Real-time monitoring (where applicable)
- Flow meters on discharge lines
- Secondary containment for treatment systems
- Sludge handling and hazardous waste disposal procedures
- Emergency spill response plan
- Host Community Development Trust, all operators must contribute 3% annual opex to this fund

3.3.2. *Monitoring & Reporting Obligations*

BFPCL in compliance with statutory requirements will implement the following:

- Monthly effluent reporting to NUPRC
- Annual Environmental Performance Report
- Environmental audit every 3 years
- Immediate reporting of non-compliance events
- Record retention for regulatory inspection

Failure to comply may attract:

- Administrative penalties
- Suspension of operations
- Environmental remediation orders

3.3.3. *Technical & Operational Verification*

To stay compliant, BFPCL like other operators in the Niger Delta must now implement:

- **Continuous Monitoring:** Real-time sensors for Oil-in-Water (OIW) content.
- **Quarterly Bioassays:** Testing the toxicity of treated water on local aquatic species (e.g., tilapia or shrimp) every 3 months.
- **Integrity Audits:** Annual pressure testing of re-injection wells to ensure produced water isn't leaking into freshwater aquifers.

3.3.4. *Regulatory Compliance Roadmap*

To ensure your project remains "audit-ready" BFPCL will implement:

- **Baseline Study:** Ensure the **Environmental Impact Assessment (EIA)** that specifically covers the chemical footprint of methanol byproducts, not just oil.
- **Water Balance Audit:** the NUPRC-required "Water Map"—showing how many barrels of water enter the plant vs. how many are recycled or disposed of.
- **Facility Commissioning:** BFPCL invite NUPRC/NMDPRA for a "*Pre-Commissioning Inspection*" of the treatment plant before starting production.

3.3.5. *Produced Water Characterization*

BFPCL will:

- Conduct full physicochemical analysis
- Determine oil-in-water (OIW) concentration
- Measure heavy metals, TDS, BTEX, PAHs, salinity, and naturally occurring radioactive materials (NORM)
- Establish baseline environmental data before discharge

Routine sampling and laboratory certification are mandatory.

3.3.6. Treatment & Processing Technologies

The project employs a multi-stage, high-fidelity treatment architecture designed to transform complex produced water into "Electronic-Grade" process feedstock.

- 1) *Primary Physical Separation & De-oiling*. Bulk hydrocarbons and suspended solids are removed via a sequential physical barrier. This stage includes **High-Capacity Slug Catchers** to manage multiphase surges from the 125km trunkline, followed by **Hydrocyclone Packages** and **Induced Gas Flotation (IGF)** units. These systems utilize centrifugal force and micro-bubble buoyancy to reduce Oil-in-Water (OIW) concentrations to <20 ppm, protecting downstream sensitive media.
- 2) *Advanced Organic Recovery & Destruction (The Organic Shield)*. To prevent catalyst carbonization ("coking") and membrane bio-fouling, the system utilizes two high-tier technologies:
 - **Methanol Stripping**: A thermal recovery column that extracts dissolved methanol and VOCs from the reaction and condensation water, recycling the product back into the synthesis loop.
 - **Advanced Oxidation Process (AOP)**: A destructive oxidation stage utilizing UV H₂O₂ to break down recalcitrant organics (BTEX and PAHs) that survive primary treatment, ensuring a Total Organic Carbon (TOC) level <0.2 ppm.
- 3) *Desalination & Mineral Polishing (The Mineral Shield)*. To achieve the near-zero conductivity required for reforming steam, a modular membrane-ion exchange hybrid is deployed:
 - **Double-Pass Reverse Osmosis (RO)**: High-pressure membrane filtration that removes 99%+ of bulk chlorides and Total Dissolved Solids (TDS).
 - **Mixed-Bed Ion Exchange (IX)**: A final demineralization stage that polishes the water to "Electronic-Grade" standards (Chlorides <0.05 ppm; Silica <0.01 ppm), essential for protecting the multi-million dollar SMR+ATR catalyst beds.
- 4) *Integrated Resource Optimization & Re-use*. Rather than simple disposal, the facility operates a **Circular Economy** loop. The treated water is converted into high-pressure steam for the **SMR + ATR Reforming Block**, meeting nearly 70% of the plant's 82 MBWPD demand internally. This minimizes freshwater withdrawal from the **St Nicholas River** and ensures the project maintains a high ESG rating while providing total disposal resilience via a **22.2km Offshore Subsea Outfall**.

3.3.7. Handling and Processing Standards

Produced water from the **500 MMscfd AKAB Hub** is characterized by high salinity, dissolved hydrocarbons (BTEX), and trace heavy metals. To transition from a waste-disposal mindset to a **Total Resource Optimization** model, the project adopts a dual-standard approach: meeting stringent regulatory effluent limits while achieving "Electronic-Grade" specifications for industrial re-use.

- 1) *Treatment Objectives & Feedstock Specifications*. The facility is designed to exceed standard EGASPIN and NMDPRA requirements. While regulatory limits focus on "safe discharge," our internal process standards focus on "catalyst protection."

Parameter	Regulatory Limit (NMDPRA/EGASPIN)	Project Inte Spec (for SMR+ATR Feed)	Strategic Rationale
Oil & Grease	<30 mg/L (Nearshore/Swamp)	Non-detectable (<0.1 mg/L)	Prevents fouling of RO membranes and coking of reformer catalysts.
Salinity (TDS)	Monitor/Ambient match	<0.1 mg/L (as NaCl)	Chlorides removed to "near-zero" to prevent irreversible catalyst poisoning.
Organics (TOC/BTEX)	Monitor	<0.2 ppm	Advanced Oxidation (AOP) destroys recalcitrant VOCs to protect synthesis purity.
Heavy Metals	<2.0 mg/L (Combined)	Trace/Removal	IX Polishing removes Lead, Mercury, & Chromium to protect high-temp alloys.

2) **Regulatory Compliance Framework.** In accordance with the Petroleum Industry Act (PIA) 2021 and recent NUPRC directives, the project implements the following handling standards:

- **Point-of-Generation Monitoring:** Real-time telemetry at the **AKAB Hub** ensures that any water not meeting the 82 MBWPD re-use specification is diverted for re-treatment rather than discharge.
- **Sensitive Ecosystem Protection:** Recognizing the proximity to the **St Nicholas River** and surrounding mangrove biodiversity, the project adopts the strictest <10 mg/L OIW standard as a baseline for the **22.2km Offshore Outfall** emergency bypass.
- **Neutralization Standard:** All waste streams from the **Ion Exchange (IX)** regeneration cycle are neutralized in a dedicated balancing tank to ensure pH stability (6.5–8.5) before reaching the subsea diffuser heads.

3.3.8. Process Integrity & Technical Handling

To ensure the treated water meets the rigorous stability requirements for the 10,000 MTPD synthesis loop, the facility implements a "Closed-Loop Integrity" protocol. This focuses on the removal of dissolved chemical constituents that standard oilfield separation cannot address.

- **Chemical Recovery (Stripping):** High-efficiency **Methanol Stripping Columns** are utilized to recover dissolved methanol and VOCs from the process and reaction water. This serves a dual purpose: maximizing product yield and reducing the **Chemical Oxygen Demand (COD)** to protect downstream membrane integrity.
- **Organic Destruction (Advanced Oxidation):** Recognizing that trace glycols (TEG/MEG) and complex hydrocarbons are resistant to standard filtration, the project utilizes **Advanced Oxidation Processes (AOP)**. By applying **UV H₂O₂** these recalcitrant chemicals are mineralized into harmless constituents, ensuring the Total Organic Carbon (TOC) remains below **0.2 ppm**.
- **Contamination Safeguard:** Unlike conventional biological treatments which risk microbial carry-over into the Reformer, the project utilizes a **sterile, non-biological physical-chemical train**. This eliminates the risk of bio-fouling in the high-temperature synthesis exchangers and ensures the >20-year design life of the SMR+ATR catalyst beds.

4. BFPCL Framework & Implementation

4.1 BFPCL Framework

The Company's Framework encompasses:

- Road Map and Decision Flow Charts,
- Benchmark & Best Practices towards compliance;
- Benchmark & Best Practices towards continuous improvement that will strengthen regulatory standing and ESG positioning:
 - Target ≤ 20 mg/L OIW (international best practice)
 - Adopt hydrocyclones + induced gas flotation systems
 - Implement reinjection where feasible (reduces surface discharge risk)
 - Deploy digital monitoring & SCADA integration
 - Conduct regular third-party lab verification
 - Align with ISO 14001 Environmental Management Systems
- Standard Operating Procedures (SOPs), including Key Performance Indicators (KPIs):
 - Development of Procedures & Methods
 - Controls for Safe Handling, and Treatment,
 - Monitoring, and Disposal of produced water generated in accordance to Petroleum Industry Act (2021)
- Key Performance Indicators (KPIs)
 - Oil-in-Water: $< 10\text{-}20$ mg/l
 - TSS: $< 5\text{-}30$ mg/l
 - ReInjection-Ready Design
- Strategic Value, including:
 - ESG Risk Reduction
 - Regulatory Compliance Alignment
 - Scalable for High Water-Cut
- Recommended Treatment Flow:
 - Separation → Hydrocyclone → IGF → DMF → [Methanol Stripper] → [AOP] → Cartridge Filter → 2-Stage RO → Ion Exchange → Reuse

4.2 BFPCL Roadmap

The Road map considers 6 concept selections (see table below) to be evaluated as part of detail design:

Industrial Re-use of the produced water has been adjudged the best option. It utilizes all the well-produced water from wet gas pushed all along the 125km, the produced water from both the 500 MMscfd Gas Processing Plant (GPP) and the 10,000 MTPD Methanol Plant (MP). This complies with the NURPC Zero Liquid Discharge (ZLD) strategy for 2026. At full operation, the water requirement for the methanol plant industrial loop will be barely met, which means zero water balance for discharge.

The features of this option include; a high-capacity finger type slug catcher at AKAB hub to manage possible slug surges and a central Water Treatment Plant (WTP) with the process flow given below.

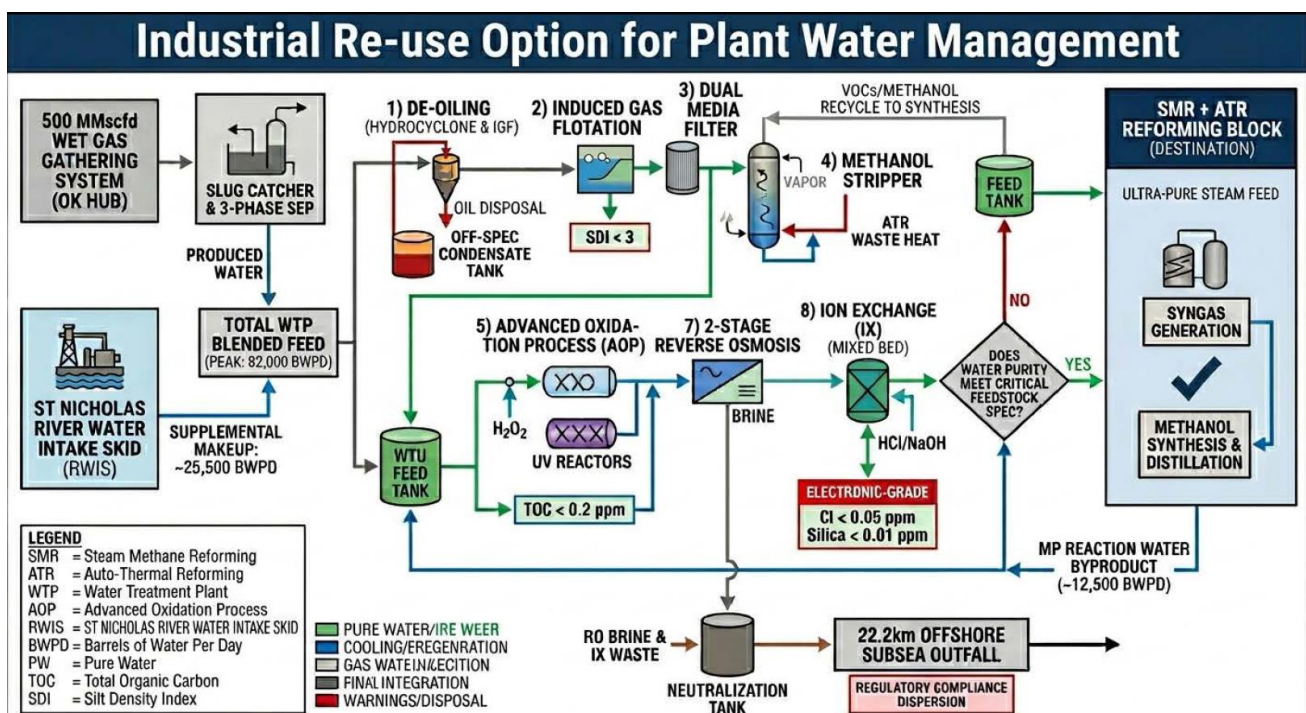
Pri- ority	Option	Key Facilities	Sub-Facilities	CAPEX (Est.)	Pros	Cons	Regulator Ease (NUPRC/NMDPRA)
1	Industrial Re-use (The Core Goal)	Central WTP & MP Utility Loop	St Nicholas RWIS, AOP, Methanol Stripper, 2-Stage RO, IX	\$68M - \$85M	Sustains 10k MTPD steam needs; Guaranteed reliability via River Intake. High ESG rating	Highest technical complexity; requires "Electronic-Grade" purity; else "Catalyst Poisoning"	Highest: Aligns with PIA 2021 Circular Economy & Green Industry mandates.
2	Hybrid: Re-use + BR Transfer	Inter-Terminal Pipeline (8")	AKAB Pumping Skid, LACT Metering, TPA Contract	\$12M - \$18M	Lowest cost synergy for Phase 1 surplus water.	Dependent on Third Party Access (TPA) and neighbor uptime.	High: Regulators favor "Facility Sharing" to minimize environmental footprint.
3	Hybrid: OKNU Source Sep + Barge	OK Hub Loading Jetty & Storage	Hydrocyclones at OKNU Hub, Double-hull Barges	\$30M - \$45M	Protects 125km trunkline from salt/corrosion at the source.	Security risks in the St Nicholas/Brass creeks; high OPEX.	High: Proactive waste management before trunkline transport.
4	Hybrid: OK Source Sep + Spiking	Multiphase Trunkline (125km)	AKAB Finger Slug Catcher (15k bbl), HP Injection Pumps	\$35M - \$40M	Minimal upstream footprint; utilizes existing gas corridor.	High corrosion risk; massive liquid handling at AKAB Terminal.	Moderate: Requires high-spec leak detection & telemetry monitoring.
5	Offshore Outfall & Emergency	22.2km Marine Pipeline	Offshore Pumps, Neutralization Tank, Diffuser Heads	\$22M - \$28M	Ultimate disposal resilience; handles RO Brine & IX waste.	Perception of discharge; requires strict NMDPRA/ECMR monitoring.	Moderate: Approved strictly as a secondary or maintenance backup.
6	3rd Party Outsourcing	On-site Reception & Loading	Transfer Pumps, Custody Meters, Contractor Vessels	\$12M - \$18M	Shifts technical/compliance burden to a contractor.	Highest per-barrel "Gate Fees"; zero control over disposal.	Low: Regulators prefer operators to own their entire waste lifecycle.

This option is combined with the Offshore Outfall option or the Transfer to Brass Terminal or barging to third party to dispose of redundant water. Excess water may occur during plant downtimes for maintenance, delayed start-up of second train or high water-cuts from wells in later years of life. The well re-injection (PWRI) option has not been considered since there are no wells available for it.

4.2.1. Process Integrated Sequence

Step	Unit Operation	Primary Function	Technical Output / Goal	Operational Risk (Failure Consequence)
1	Slug Catcher & 3-Phase Sep	Primary gravity separation of Gas, Oil, and Water.	Stabilized Feed: Removal of 99% bulk liquids.	Slug Overload: Liquid carry-over into GPP, causing immediate plant trip.
2	De-oiling (Hydrocyclone / IGF)	Centrifugal & buoyancy-driven oil removal.	OIW < 20 ppm: Protection of downstream filter media.	Media Fouling: Oil coats filters and RO membranes, requiring total replacement.
3	Filtration (DMF/Cartridge)	Depth filtration and 1–5 micron safety barrier.	SDI < 3: Elimination of physical particulates / silt.	Membrane Abrasion: Silt/sand punctures RO membranes, leading to salt passage.
4	Methanol Stripper	Thermal recovery of Methanol and VOCs.	Product Recovery: Prevention of RO bio-fouling.	Bio-fouling: Organic "food" creates bacterial slime, plugging the RO system.
5	Advanced Oxidation (AOP)	UV + H ₂ O ₂ destruction of trace organics.	TOC < 0.2 ppm: Elimination of BTEX/PAHs.	Catalyst Coking: Trace organics "crack" into soot, destroying Re-former catalysts.
6	Reverse Osmosis (2-Pass)	Molecular desalination via semi-permeable membranes.	99% TDS Removal: Primary mineral reduction.	Scaling: Mineral buildup (Calcium/Silica) on membranes, reducing water flow.
7	Ion Exchange (Mixed Bed)	Final polishing of residual ions/minerals.	"Electronic-Grade": Cl < 0.05 ppm; Silica < 0.01 ppm.	Catalyst Poisoning: Trace Chlorides cause irreversible chemical deactivation of ATR/SMR.
8	SMR + ATR Reforming Block	Catalytic conversion of dry gas + ultra-pure steam.	Stoichiometric Syngas: Target H ₂ /CO ratio.	Production Loss: Low-quality steam causes emergency shutdown of 10,000 MTPD plant.
9	Neutralization & Outfall	pH balancing of RO Brine and IX waste.	Regulatory Compliance: Meeting NMDPRA standards.	Regulatory Fine: Discharge of aggressive waste leads to environmental penalties/fines.

4.2.2. Plant Water Management Decision Flow Chart



5. Produced Water Characterisation & Risk Assessment

5.1 Background Information

5.1.1. Source & Composition

Produced water typically contains:

- High salinity and dissolved solids
- Hydrocarbons (oil & grease), organic compounds
- Heavy metals (Fe, Pb, Ni, Cr, etc.)
- Other inorganic constituents

These contaminants vary by field and over time and often exceed regulatory limits without treatment.

5.1.2. Environmental & Health Risk Assessment

- **Toxicity to ecosystems:** Untreated produced water can harm aquatic life and soil quality, affecting fisheries and agriculture.
- **Human exposure risks:** Communities relying on surface/groundwater can be adversely affected if produced water is improperly discharged.

5.1.3. Data & Monitoring Plan

- **Baseline sampling:** Establish pre-production water quality baselines.
- **Routine testing:** Frequent sampling of produced water for parameters like oil & grease, TDS, metals, BOD/COD.
- **Trend analysis:** Use time-series data to detect deterioration or spikes in contaminants.

5.2 Treatment & Management Strategies

A robust produced water management strategy combines **prevention, treatment, reuse, and disposal**.

5.2.1 Source Reduction & Production Practices

- **Downhole oil-water separation:** Separates water before reaching surface facilities, reducing downstream handling volumes.
- **Reservoir management:** Reduce water cut through well interventions and pressure maintenance to minimize produced water volumes.

5.2.2 On-Site Treatment Technologies

A layered approach combining physical, chemical, and biological treatment:

- **Primary separation:** Gravity separators or Gas-Oil Separation Plants (GOSP) remove bulk oil and water phases.
- **API separators:** Used for initial removal of free oil and suspended solids.
- **Induced gas flotation & skimming:** Enhance removal of dispersed oil droplets.

- **Chemical dosing:** Demulsifiers, coagulants and flocculants assist aggregation of pollutants.
- **Adsorption & filtration:** Media filters (e.g., walnut shell filters), bio-adsorbents (groundnut shells, sugarcane bagasse, etc.) can reduce metals and residual hydrocarbons.
- **Biological treatment:** Emerging techniques using microbes or algae can reduce organic loads.

5.2.3 *Off-Site / Advanced Options*

- **Reinjection for disposal or EOR:** Treated water can be reinjected into subsurface formations for disposal or reservoir pressure support.
- **Zero Liquid Discharge (ZLD) approaches:** Advanced evaporation/desiccation technologies can eliminate liquid discharge and generate solid by-products.

5.3 Water Reuse & Resource Optimization

5.3.1 *Internal Reuse*

- **Operational reuse:** Use treated produced water for drilling, work-overs, or pressure maintenance where permitted.
- **Reduced freshwater demand:** Prioritize reuse to mitigate freshwater consumption, especially in water-scarce zones.

5.3.2 *Beneficial Reuse Opportunities*

- **Agricultural irrigation or industrial uses:** Only if water quality meets safe reuse thresholds and community consent.
- **Community supply potential:** Explore long-term treatment pathways that could support local water demands, subject to stringent safety standards.

5.4 Monitoring, Reporting & Compliance Assurance

5.4.1 *Environmental Monitoring*

- Continuous monitoring systems at discharge points.
- Performance checks for treatment systems.
- Independent audits and third-party verification where appropriate.

5.4.2 *Reporting*

- Periodic compliance reports to NUPRC & environmental regulators.
- Public transparency on water quality and management performance.

5.4.3 *Corrective Actions*

- **Non-compliance triggers** mandatory review and mitigation plans.
- **Incident response plans** in case of spills or accidental discharges.

5.5 Stakeholder Engagement & Community Safeguards

5.5.1 *Community Consultation and HCDT Framework*

- Regular engagement with host communities on environmental concerns, water impacts, and management plans.
- The project must establish a Host Community Development Trust (HCDT) via the [HostComply](#) portal.
- Funding: 3% of the actual annual Operating Expenditure (OPEX).

5.5.2 *Grievance Mechanisms*

- Establish clear channels for communities to report water quality issues or grievances.
- Grievance Redress: A formal tracker must be maintained; environmental grievances must be acknowledged within 48 hours.

5.5.3 *Capacity Building*

- Support local training and education on produced water risks and safeguards
- .Skills Acquisition: Training curriculum must be OGTAN-certified and registered on the NCDMB NOGIC JQS portal. Key modules include Pipeline Surveillance and Methanol Safety.

5.6 [Research, Innovation & Continuous Improvement](#)

5.6.1 *Technology Trials*

Pilot novel treatment and reuse technologies, including bio-adsorbents and advanced separation systems.

5.6.2 *Standards Review*

Regularly update local discharge and reuse criteria to align with **international best practices** and emerging science.

5.6.3 *Collaborative Initiatives*

Industry-government partnerships for data sharing, research and sustainable water management strategies.

5.7 [Summary of Best Practices](#)

- 1) **Strict compliance with environmental regulations and effluent limits.**
- 2) **Comprehensive characterization of produced water** to inform treatment design.
- 3) **Implementation of layered treatment systems** combining physical, chemical, and biological methods.
- 4) **Promotion of reuse and resource optimization** to reduce freshwater demand.

- 5) **Robust monitoring, reporting, and adaptive management** to ensure long-term compliance.
- 6) **Meaningful stakeholder engagement**, especially with local communities.

6. Produced Water Management Policy

6.1 Scope

This policy applies to:

- All onshore, swamp, and shallow offshore assets in the Niger Delta
- Flow stations, central processing facilities (CPF), early production facilities (EPF)
- Produced water handling, storage, treatment, reuse, discharge, and reinjection systems
- Contractors and third-party operators

6.2 Discharge Standards

Produced water shall not be discharged unless it meets prescribed limits for:

- Oil & Grease
- Total Suspended Solids (TSS)
- Total Dissolved Solids (TDS)
- pH
- Heavy Metals (Fe, Pb, Cr, Ni, Cd, etc.)
- Temperature

Stricter limits apply for inland/swamp discharge compared to offshore discharge.

6.3 Produced Water Management Hierarchy

The Company adopts the following hierarchy (highest priority first):

6.3.1 Prevention & Minimization

- Reservoir management to reduce water cut
- Well interventions to control water production

6.3.2 Reuse & Reinjection

- Pressure maintenance
- Enhanced Oil Recovery (EOR)
- Operational reuse (drilling/workover fluids)

6.3.3 Treatment & Controlled Discharge

- Multi-stage treatment systems
- Continuous monitoring before discharge

6.3.4 Disposal

- Approved subsurface disposal wells
- No uncontrolled surface discharge

6.4 Technical Standards

6.4.1 Characterization

Produced water shall be characterized for:

- Hydrocarbon content
- Salinity
- Metals
- Naturally Occurring Radioactive Materials (where applicable)
- Dissolved gases

Baseline environmental data must be established before production.

6.4.2 *Treatment Philosophy*

A multi-barrier treatment approach shall be implemented:

Stage 1 – Primary Separation

- Three-phase separators
- Heater treaters
- Free water knockouts

Stage 2 – Secondary Treatment

- Hydrocyclones
- Induced Gas Flotation (IGF) units
- Skim tanks

Stage 3 – Tertiary/Polishing

- Media filters (e.g., walnut shell filters)
- Cartridge filters
- Adsorption systems
- Chemical polishing

Discharge is permitted only after compliance verification.

6.5 **Monitoring & Reporting**

6.5.1 *Monitoring Requirements*

- Daily operational monitoring
- Weekly laboratory analysis (minimum)
- Continuous Oil-in-Water analyzer at discharge point
- Quarterly third-party validation

6.5.2 *Reporting*

- Monthly compliance reports to NUPRC
- Immediate reporting of exceedances
- Annual environmental performance report

6.5.3 *Risk Management & Incident Response*

- Produced water spill response plan aligned with NOSDRA requirements
- Secondary containment for storage tanks
- Emergency shutdown systems
- Root cause analysis for any exceedance

6.6 Community & Environmental Safeguards

- Stakeholder engagement prior to discharge approvals
- Public grievance mechanism
- Protection of wetlands, mangroves, fisheries, and groundwater
- No discharge into potable water sources

6.7 Governance & Accountability

S/N	Role	Responsibility
1.	Asset Manager	Overall compliance
2.	Production Superintendent	Operational integrity
3.	Environmental Manager	Monitoring & reporting
4.	HSE Team	Audit & training
5.	Contractors	Compliance with this policy

6.8 Continuous Improvement

- Annual performance review
- Technology upgrades evaluation
- Water reuse expansion targets
- Benchmarking against international best practice

6.9 The Linear Flow Description:

The process follows a high-fidelity linear progression: Primary Gravity Separation is followed by centrifugal de-oiling (Hydrocyclone) and flotation (IGF). The stream then undergoes depth filtration (DMF) before entering the thermal recovery phase (Methanol Stripping) and the chemical destruction phase (AOP). Final polishing is achieved through high-rejection 2-Stage Reverse Osmosis and Mixed-Bed Ion Exchange, delivering 'Electronic-Grade' water to the Synthesis Block.

7. PRODUCED WATER VOLUME MANAGEMENT

7.1 Produced Water Volumes

The total produced water volume is given in the table below:

Source	Volume (BWPD)	Type
Wet Gas Wells	36,000-40,000	Formation Brine (Saline)
GPP Condensation	2,500 – 4,000	High Purity Condensate
MP Reaction Water	9,500 – 12,500	Process Byproduct (Organic-rich)
TOTAL INT SUPPLY	52,000 – 56,500	Total available for Re-use
Total Consumption	82,000	Ultra-Pure Steam
Water Gap	25,500	River Water Intake

7.1.1 Peak Water Consumption Analysis

For a 10,000 MTPD plant using Combined Reforming (SMR + ATR), the peak water demand is driven by the Steam-to-Carbon (S/C) ratio.

- Peak Demand: ~1.3 tons of high-purity water per ton of Methanol.
- Daily Metric Volume: 13,000 Metric Tons/Day.
- Volumetric Equivalent: ~82,000 Barrels Per Day (BPD).

Parameter	Current	Peak Forecast
Average Daily	36,000 bwpd	40,000 BWPD
Design Capacity	52,000 BWPD	—

System designed with $\pm 30\%$ contingency capacity.

7.2 River Water Intake

Regulatory Implications of the "Water Gap"

Since you now require supplemental water, your regulatory burden expands beyond disposal:

- **River Water Intake Permit:** Required from the NIWA (National Inland Waterways Authority) and the NMDPRA.
- **Water Footprint Reporting:** The volume of water extracted from the St Nicholas River vs. the volume of Produced Water re-used must now be reported in the ECMR.
- **Brine Management:** Even with 82,000 BPD consumption, the RO Brine (~20,000 BWPD) and IX Waste (~800 BPD) must still be discharged via the Off-shore Outfall.

7.3 Strategic FDP Justification

The water management strategy for the 500 MMscfd Bayelsa Hub is engineered for **Total Resource Optimization**, utilizing a high-efficiency **Two-Step Reforming (SMR + ATR)** configuration that requires a peak ultra-pure steam load of **82 MBWPD**. The project leverages the gas gathering network as a primary 'water mine,' providing **~56,5 MBWPD** of internal supply (well brine, condensation, and reaction water) supplemented by a dedicated **St Nicholas River Intake Skid** to guarantee feedstock reliability regardless of well-water fluctuations. This internal supply is secured by a 'Triple-Shield' treatment strategy—integrating **Methanol Stripping, Advanced Oxidation (AOP), and Ion Exchange**—to destroy trace organics (BTEX) and achieve 'electronic-grade' purity. This technical 'insurance' ensures maximum protection for multi-million dollar synthesis catalysts and extends the operational lifespan of the facility to 20+ years, while the **22.2km Offshore Outfall** provides the ultimate disposal resilience to maintain uninterrupted gas production under all operational scenarios.

7.4 Treatment Philosophy & Process Design

7.4.1 Treatment Objectives

- Reduce OIW to ≤ 20 mg/L
- Remove suspended solids
- Control scaling and corrosion
- Enable safe reinjection

7.4.2 Process Description

Produced Water Treatment Integrated Sequence:

- 1) **Three-Phase Separator**: Primary gravity separation of gas, liquid hydrocarbons, and formation brine from the OKNU-BBOU-ELEPA-AKAB gathering lines.
- 2) **Hydrocyclone Package**: Centrifugal de-oiling to reduce bulk oil-in-water (OIW) concentrations.
- 3) **Induced Gas Flotation (IGF)**: Polishing stage using gas-bubble buoyancy to remove residual emulsified oil.
- 4) **Dual Media Filters (DMF)**: Depth filtration to remove suspended solids, protecting the downstream stripping and oxidation units.
- 5) **Methanol Stripper**: Thermal recovery unit that extracts dissolved methanol from the MP Reaction Water and GPP Condensate for recycle, while removing volatile organics from the well water.
- 6) **Advanced Oxidation Process (AOP)**: Integration of UV and H_2O_2 to destroy recalcitrant organics (BTEX/PAHs) that survive stripping, protecting the RO membranes from bio-fouling.
- 7) **Cartridge Polishing Filters**: Fine safety barrier to ensure a Silt Density Index (SDI) < 3.0 .
- 8) **Double Pass Reverse Osmosis (RO)**: Bulk desalination stage removing 99%+ of bulk Chlorides and Total Dissolved Solids (TDS).
- 9) **Ion Exchange (IX)**: Mixed-bed polishing to achieve "electronic-grade" purity (Chlorides < 0.05 ppm).
- 10) **SMR + ATR Reforming Block**: The destination for the treated 82,000 bwpd stream. The water is converted to high-pressure steam and reacted with the 500 MMscfd gas feed to produce Syngas.

7.4.3 Chemical Treatment

- Demulsifier
- Corrosion inhibitor
- Scale inhibitor
- Biocide (controlled dosage)

All chemicals approved and MSDS attached.

7.4.4 Monitoring Systems

- Online Oil-in-Water analyzer
- Flow meters
- Pressure indicators
- High-OIW alarm & automatic diversion to slop tank

7.5 Disposal Strategy

7.5.1 Industrial Re-use

Produced water will be recycled for industrial use by the Methanol Plant.

Process sequence will follow section 7.2.2 above

7.5.2 Contingency Discharge

If water is excess of Industrial Re-use;

- Treated water discharged to controlled Offshore outfall
- Discharge point coordinates documented
- OIW continuously monitored
- Immediate shutdown if limits exceeded

7.6 Compliance With Discharge Limits

System designed to meet:

- Oil in Water: ≤ 29 mg/L
- No visible sheen
- TSS within EGASPIN limits
- Heavy metals within allowable thresholds

Design target OIW: ≤ 15 mg/L (internal standard)

7.7 Environmental Risk Assessment

HAZOP conducted on produced water system.

Identified risks:

- Separator upset
- IGF failure
- Pipeline rupture
- Sludge overflow

Mitigation measures:

- Redundant pumps
- Secondary containment
- Automated shutdown
- Spill kits at discharge point

Environmental sensitivity assessment confirms proximity to mangrove ecosystem; therefore, reinjection prioritized.

7.8 Monitoring & Reporting

7.8.1 Sampling Frequency

- OIW: Continuous monitoring
- Full lab analysis: Monthly
- Heavy metals: Quarterly

7.8.2 Reporting

- Monthly compliance report to NUPRC
- Annual Environmental Performance Report
- Immediate notification to NUPRC and NOSDRA in case of exceedance

7.9 Sludge & Waste Management

Estimated sludge generation: 1.2 m³/day.

Management:

- Stored in closed sludge tank
- Removed quarterly
- Disposed via licensed waste contractor
- Waste manifest documentation maintained

7.10 Integrity Management

- Corrosion coupons installed
- Chemical dosing optimization
- Quarterly pipeline inspection
- Cathodic protection system installed

7.11 Emergency Response

Integrated with Oil Spill Contingency Plan.

In event of exceedance:

- 1) Immediate diversion
- 2) Shut down discharge
- 3) Notify NUPRC within regulatory timeframe
- 4) Root cause investigation

7.12 Training & Governance

- Environmental Officer appointed

- Operators trained quarterly
- Competency matrix maintained
- Internal audit conducted annually

7.13 Decommissioning & Abandonment

At end-of-field, Decommissioning & Abandonment will follow the table below;

S/N	Asset	Removal Strategy
1	Offshore Outfall (22km)	Flushing & Retrieval.
2	125km OKNU-AKAB Trunkline	Cleaning & In-situ Capping.
3	Water Treatment Plant (WTP)	Dismantling & Remediation.
4	River Water Intake Skid	Dismantling & Remediation
5	Gas Processing Plant (GPP)	Dismantling & Remediation.
6	Methanol Plant (MP)	Dismantling & Remediation.
7	Brass Export Terminal	Dismantling & Remediation.

Final environmental closure report submitted

8. CONCLUSION

BFPCL confirms that produced water generated at OML 33 will be:

- Properly characterized
- Adequately treated
- Preferentially recycled in Industrial Re-use to achieve Zero Liquid Discharge (ZLD)
- Continuously monitored
- Fully compliant with NUPRC regulatory requirements

The system will be designed with environmental protection as the primary objective in this sensitive Niger Delta swamp environment.